

Set It and Forget It

Secure & Automated Certificate Management on AKS

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Agenda



Cert-Manager

Let's Encrypt

Domain Ownership Validation

Demo

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Cert-Manager

Cert-Manager

Automatically creates/updates certificates

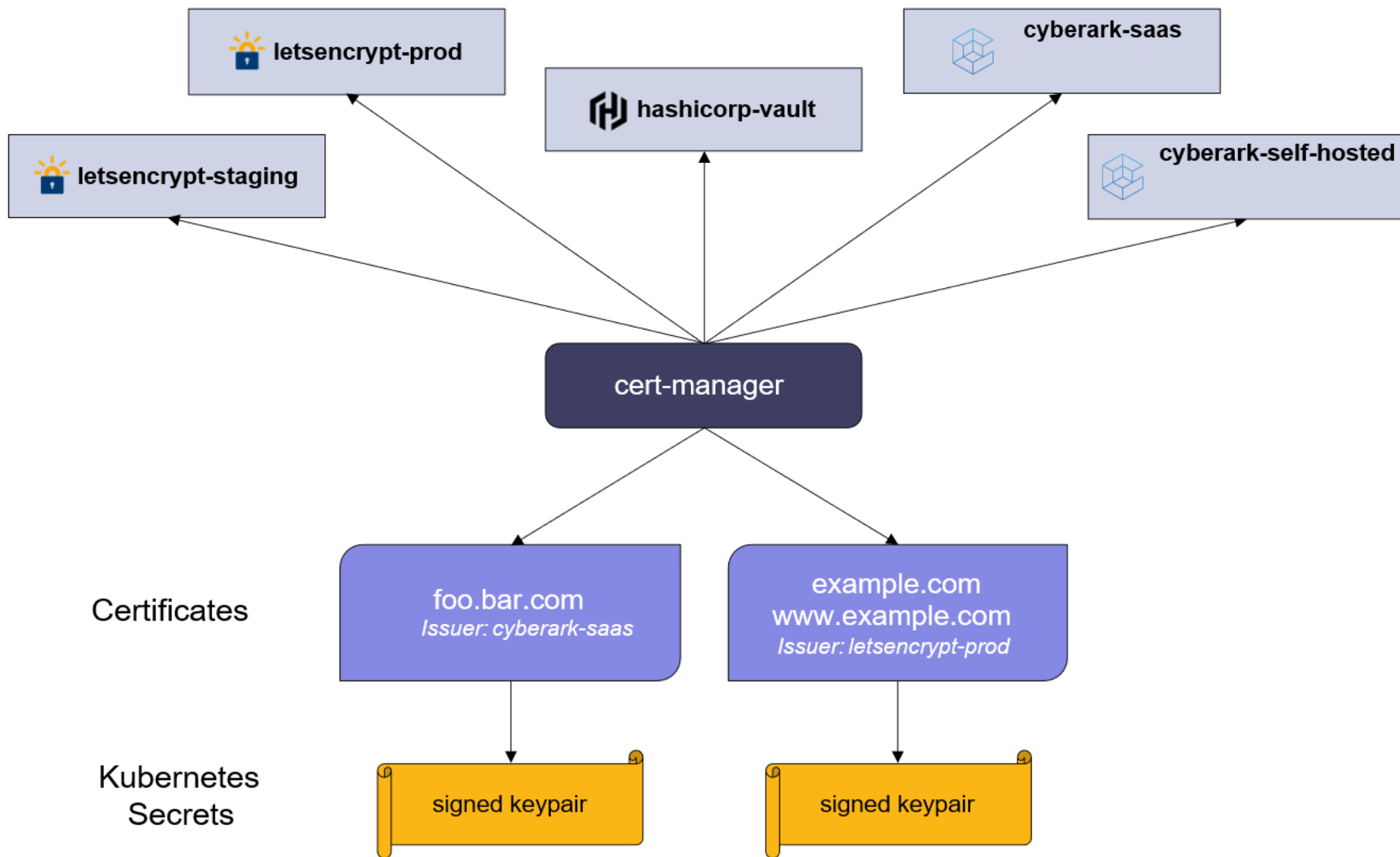
Uses Issuer to get certificates

Supports Ingress and Gateway API

Verifies ownership of Domain



Issuers



Let's Encrypt

Let's Encrypt

Nonprofit to create a more secure internet

Free certificates, valid for 90 days

Staging and Production environment

Prod certificates are fully valid in modern
browsers



Domain Ownership Validation

Domain Ownership Validation

HTTP-01 challenge

```
solvers:
  - http01: 
      gatewayHTTPRoute:
        parentRefs:
          - name: $GatewayName
            namespace: $GatewayNamespace
            kind: Gateway
```

```
solvers:
  - http01:
      ingress:
        class: nginx
```

```
solvers:  
  - http01:  
    gatewayHTTPRoute:  
      parentRefs:  
        - name: $GatewayName  
          namespace: $GatewayNamespace  
          kind: Gateway
```

```
solvers:  
  - http01:  
    ingress:  
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solvers:
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solvers:  
  - http01:  
    gatewayHTTPRoute:  
      parentRefs:  
        - name: $GatewayName  
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          kind: Gateway
```

```
solvers:  
  - http01:  
    ingress:  
      class: nginx
```

Domain Ownership Validation

HTTP-01 challenge

Proof ownership of a domain with a file

`http://<YOUR_DOMAIN>/.well-known/acme-challenge/<TOKEN>`

Cannot be used for wildcard certificates

Domain Ownership Validation

DNS-01 challenge

- dns01: ←

azureDNS:

hostedZoneName: \$DnsZoneName

resourceGroupName: \$DnsResourceGroup

subscriptionID: \$DnsSubscriptionId

environment: AzurePublicCloud

managedIdentity:

clientID: \$CertManagerManagedIdentityClientId

Domain Ownership Validation

DNS-01 challenge

Proof ownership of a domain with DNS record

_acme-challenge.<YOUR_DOMAIN>

Challenge can be delegated to another DNS zone

Demo

Demo

Deploy new AKS cluster

Use Envoy Gateway API

Deploy Cert-Manager

Create ClusterIssuer with HTTP-01 Challenge

Deploy Demo app to test HTTPs connection

Demo

Create and configure Managed Identity for DNS access

Update Cert-Manager to use Entra Workload ID

DNS-01 Challenge

Test wildcard certificate

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